

Part 1: inputs

m = # of training images.

classes = # of possible ways to classify the input. (0-9)



28px

Total number of pixels is 784px

28px

X = Dataset

$$X = \begin{bmatrix} - & x^0 & - \\ - & x^1 & - \\ - & \vdots & - \\ - & x^m & - \end{bmatrix}$$

; We need to transpose this

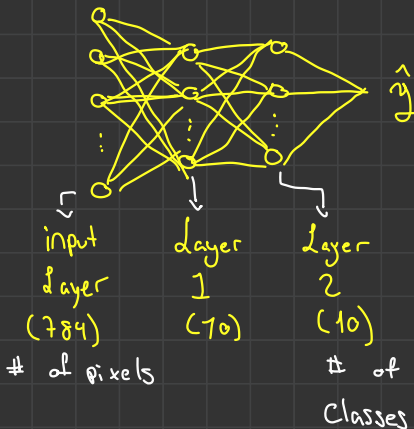
Why?

$$X^T = \begin{bmatrix} | & | & \dots & | \\ x^0 & x^1 & \dots & x^m \\ | & | & \dots & | \end{bmatrix}$$

of pixels =

of input neurons

Part 2: neural network



Binary Classification has 1 neuron?

Part 3: Forward Propagation

Forward Propagation = We input an image to get a prediction out of it (no learning)

Input layer.

$$\underline{A^0} = X \rightarrow 284 \times m$$

W = Weight

b = Bias

Unactivated layer 1:

$$\underline{z^1} = \underset{10 \times m}{W^1} \underset{284 \times m}{A^0} + \underset{10 \times 1}{b^1}$$

Activated Layer 1:

$$\underline{A^1} = g(\underline{z^1}) \rightarrow \text{ReLU}(\underline{z^1})$$

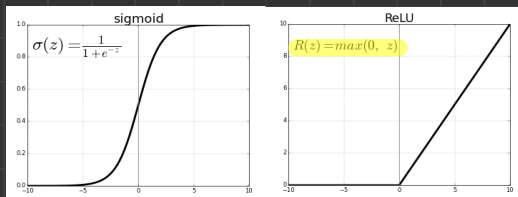
Unactivated Layer 2:

$$\underline{z^2} = \underset{10 \times m}{W^2} \underset{10 \times 1}{A^1} + \underset{10 \times 1}{b^2}$$

Activated Layer 2:

$$\underline{A^2} = g(\underline{z^2}) \rightarrow \text{Softmax}(\underline{z^2})$$

Activation functions



Part 3: Back Propagation

Back Propagation = From the prediction and ground truth, calculate the error and adapt weights.

Back Prop Layer 2:

$$\underline{\delta z^2} = \underset{10 \times m}{A^2} - \underset{10 \times m}{y} \quad (\text{get error})$$

y = One-hot
encoded truth
 $\begin{bmatrix} 0 \\ 1 \\ 0 \\ \vdots \end{bmatrix}$

How much b^2 and W^2 Contributed:

$$\underline{\delta W^2} = \frac{1}{m} \underline{\delta z^2} \underline{A^{1T}}$$

Same thing for layer #1

$$\underline{\delta z^1} = \underline{w^{21}} \underline{\delta z^2} \cdot g'(\underline{z^1})$$

$$\underline{\delta w^1} = \frac{1}{m} \underline{\delta z^1} x^T$$

$$\underline{\delta b^1} = \frac{1}{m} \sum \underline{\delta z^1}$$

Update Parameters :

$$w^1 = w^1 - \alpha \underline{\delta w^1}$$

$$b^1 = b^1 - \alpha \underline{\delta b^1}$$

$$w^2 = w^2 - \alpha \underline{\delta w^2}$$

$$b^2 = b^2 - \alpha \underline{\delta w^2}$$

α = learning rate, AKA size of steps to achieve low error (convergence)